AMC 12A 2023 Problem 15

Strategic Analysis

Using Framework v2.1

Problem Statement

Usain is walking for exercise by zigzagging across a 100-meter by 30-meter rectangular field, beginning at point A and ending on the segment $\overline{BC}$. He wants to increase the distance walked by zigzagging as shown in the figure. What angle $\theta = \angle PAB = \angle QPC = \angle RQB = \cdots$ will produce a length that is 120 meters?

(A) $\arccos \frac{5}{6}$ **(B)** $\arccos \frac{4}{5}$ **(C)** $\arccos \frac{3}{10}$ **(D)** $\arcsin \frac{4}{5}$ **(E)** $\arcsin \frac{5}{6}$

Note: The figure shows a rectangular field $ABCD$ with A at bottom-left, B at bottom-right, C at top-right, D at top-left. The zigzag path goes from A to P (on $\overline{DC}$), then to Q (on $\overline{AB}$), then to R (on $\overline{DC}$), and so on, with all angles θ equal.

Problem Configuration Analysis

1. Problem Configuration Analysis

A. Problem Classification

- **Competition**: AMC 12A 2023
- **Problem Number**: 15
- **Difficulty Band**: Mid-range (Problem 15/25)
- **Expected Time**: 3-4 minutes
- **Point Value**: 6 points (standard AMC 12 scoring)

B. Mathematical Domain Classification

- **Primary Domain**: Geometry / Trigonometry
- **Secondary Domain**: Path Optimization / Reflection
- **Tertiary Domain**: Similar Triangles / Pythagorean Theorem
- **Cross-Domain Potential**: High (connects geometry, trigonometry, physics concepts)

C. Problem Type

- **Format**: Multiple choice (angle determination)
- **Question Type**: "What angle..." (inverse trigonometric function)
- **Core Task**: Find angle θ that produces a specific path length
- **Answer Form**: Inverse trigonometric expression

D. Given Information

- Rectangular field dimensions: 100 m (length) $\times$ 30 m (width)
- Starting point: A (corner of rectangle)
- Ending location: somewhere on segment $\overline{BC}$
- All zigzag angles are equal: $\theta = \angle PAB = \angle QPC = \angle RQB = \cdots$
- Total path length: 120 meters
- The number of zigzag segments is not specified

E. Constraints

- Path must zigzag between top and bottom edges
- All angles θ must be equal
- Path starts at A and ends on $\overline{BC}$
- Total length must be exactly 120 meters
- Field dimensions are fixed

F. Objective

- **Find**: The angle θ that produces a 120-meter path
- **Output**: Select the correct inverse trigonometric expression

G. Key Structural Features

- **Reflection property**: Zigzag path resembles light reflecting between mirrors
- **Unfolding potential**: Path can be "unfolded" into a straight line
- **Similar triangles**: All zigzag segments form similar right triangles
- **Constant ratio**: Horizontal distance : path length $= 100 : 120 = 5 : 6$
- **Physics analogy**: Horizontal velocity component is constant

H. Strategic Orientation

- **Primary Strategy**: Unfolding technique (flatten zigzag into straight line)
- **Key Insight**: $\cos\theta$ represents horizontal progress per unit path length
- **Expected Difficulty**: Moderate - requires geometric insight or trigonometric setup
- **Time Pressure**: Should be completed in 3-4 minutes

Strategic Framework Application

2. Strategic Framework Application

A. Psychological Strategies

- **Confidence Calibration**:
  - Mid-range problem with elegant solution
  - Multiple solution approaches available
  - Answer choices help narrow possibilities
- **Pattern Recognition**:
  - Recognize zigzag/reflection pattern
  - Connection to unfolding technique
  - Ratio $100 : 120 = 5 : 6$ suggests $\cos\theta = 5/6$
- **Problem Familiarity**:
  - Similar to light reflection problems
  - Related to path optimization
  - Common geometry/trig problem type

B. Investigation Strategies

- **Initial Exploration**:
  - Observe that horizontal distance is 100 m, path length is 120 m
  - Ratio: $\frac{100}{120} = \frac{5}{6}$
  - This suggests $\cos\theta = 5/6$ (horizontal component)
- **Visualization Techniques**:
  - Draw one zigzag segment with its projection
  - Imagine unfolding the path into a straight line
  - Consider the physics analogy: constant horizontal velocity
- **Multiple Approaches**:

- Method 1: Unfolding (most elegant)
- Method 2: Similar triangles and proportions
- Method 3: Algebraic setup with Pythagorean theorem
- Method 4: Physics/velocity decomposition

C. Argument Construction

- **Logical Structure (Unfolding Method)**:
  1. Visualize unfolding the zigzag path
  2. When unfolded, path becomes a straight line of length 120 m
  3. Horizontal distance remains 100 m
  4. In the right triangle formed: $\cos\theta = \frac{100}{120} = \frac{5}{6}$
  5. Therefore: $\theta = \arccos\frac{5}{6}$
- **Key Deductions**:
  - Horizontal progress per unit path length $= \cos\theta$
  - Total horizontal progress = (total path length) $\times \cos\theta$
  - Therefore: $120\cos\theta = 100$
  - So: $\cos\theta = 5/6$

D. Cross-Domain Translation

- **Geometric Interpretation**: Unfolding creates a right triangle
- **Trigonometric Interpretation**: $\cos\theta$ is the horizontal component
- **Physics Interpretation**: Horizontal velocity component is constant
- **Algebraic Interpretation**: Solve for θ using constraint equation

Tactical Selection

3. Tactical Methodology Selection

A. Core Mathematical Tactics

- **Unfolding/Reflection**: Flatten zigzag path into straight line
- **Similar Triangles**: All zigzag segments are similar
- **Trigonometric Ratios**: Use $\cos\theta$ for horizontal component
- **Ratio Analysis**: Compare horizontal distance to path length
- **Pythagorean Theorem**: Relate segment lengths (alternative approach)

B. Domain-Specific Tactics

- **Geometry**:
  - Unfolding preserves path length but changes configuration
  - Reflection angles create equal angles
  - Similar triangles have proportional sides
- **Trigonometry**:
  - $\cos\theta$ represents horizontal progress rate
  - Inverse trig functions recover angle from ratio
  - Answer choices indicate arccos vs arcsin
- **Problem-Solving Techniques**:
  - Draw auxiliary lines (projections)
  - Use physics intuition (velocity components)
  - Check answer choices for plausibility

C. Tactical Priorities

1. Recognize the $100 : 120 = 5 : 6$ ratio (30 seconds)
2. Connect ratio to $\cos\theta$ (45 seconds)
3. Visualize unfolding or similar triangles (60 seconds)
4. Set up equation: $\cos\theta = 5/6$ (30 seconds)
5. Identify answer: $\arccos(5/6)$ (30 seconds)
6. Verify with answer choices (30 seconds)

Conversion Techniques

4. Conversion Techniques Application

A. Method 1: Unfolding (Most Elegant)

- **Concept**: Imagine "unfolding" the zigzag by reflecting alternating segments
- When unfolded, the path becomes a straight line from A to some point S'
- The unfolded path has:
  - Horizontal distance: 100 m (same as rectangle length)
  - Total path length: 120 m (given)

- This creates a right triangle where:

$$\begin{aligned}\cos\theta &= \frac{\text{adjacent (horizontal)}}{\text{hypotenuse (path)}} \\ &= \frac{100}{120} \\ &= \frac{5}{6}\end{aligned}$$

- Therefore: $\theta = \arccos\frac{5}{6}$

B. Method 2: Similar Triangles and Proportions

- Drop perpendicular from P to $\overline{AB}$, meeting at point Q'
- Let $AQ' = x$, then $Q'P = 30$ (width of rectangle)
- By Pythagorean theorem: $AP = \sqrt{x^2 + 900}$
- All zigzag segments are similar, so they're all proportional
- Total horizontal distance = sum of all horizontal projections = 100
- Total path length = sum of all segment lengths = 120
- The ratio is preserved: if horizontal is x, path is $\frac{6x}{5}$
- Therefore: $\cos\theta = \frac{x}{\frac{6x}{5}} = \frac{5}{6}$

C. Method 3: Algebraic Setup

- Let x = horizontal projection of one full segment
- Then the hypotenuse of one full segment $= \sqrt{x^2 + 900}$
- Number of segments $= \frac{100}{x}$ (may not be integer, but proportions work)
- Total path length:

$$\frac{100}{x} \cdot \sqrt{x^2 + 900} = 120$$

- Simplify:

$$\sqrt{x^2 + 900} = \frac{6x}{5}$$

- Square both sides:

$$\begin{aligned}x^2 + 900 &= \frac{36x^2}{25} \\ 25x^2 + 22500 &= 36x^2 \\ 11x^2 &= 22500 \\ x^2 &= \frac{22500}{11}\end{aligned}$$

- Then:

$$\cos\theta = \frac{x}{\sqrt{x^2 + 900}} = \frac{x}{\frac{6x}{5}} = \frac{5}{6}$$

D. Method 4: Physics Analogy

- Imagine Usain walks at constant speed
- Horizontal velocity component is constant (like light reflecting off mirrors)
- Horizontal velocity = (total velocity) $\times \cos\theta$
- Time is proportional to distance walked
- Horizontal distance = (horizontal velocity) $\times$ (time)
- So: horizontal distance = (total distance) $\times \cos\theta$
- Therefore: $100 = 120\cos\theta$
- Thus: $\cos\theta = \frac{100}{120} = \frac{5}{6}$

E. Answer Selection

- We found $\cos\theta = \frac{5}{6}$
- Therefore $\theta = \arccos\frac{5}{6}$
- Check answer choices: This is **(A)**
- Verify it's not arcsin: $\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{25}{36}} = \sqrt{\frac{11}{36}} = \frac{\sqrt{11}}{6} \neq \frac{5}{6}$

Verification Strategy

5. Verification Strategy Design

A. Verification Level

- **Selected Level**: Level 3 - Reasonableness Check + Alternative Method
- **Reasoning**: Geometric problem with elegant solution
- **Time Budget**: 45-60 seconds

B. Verification Checklist

- **Check ratio**:
  - $\frac{100}{120} = \frac{5}{6}$ ✓
  - This matches $\cos\theta = 5/6$
- **Verify angle type**:
  - Since $\cos\theta = 5/6 < 1$, angle exists ✓
  - Since $5/6 > 0$, angle is acute ✓
  - Answer should be arccos, not arcsin ✓

- **Check answer format**:
  - Answer (A): $\arccos \frac{5}{6}$ matches our result ✓
- **Reasonableness**:
  - Path length $120 > 100$ (straight distance) makes sense ✓
  - Zigzag adds 20% extra distance, reasonable ✓

C. Alternative Verification Methods

- **Method 1**: Compute $\theta \approx \arccos(0.833) \approx 33.6°$, seems reasonable for zigzag
- **Method 2**: Check that $\sin\theta = \frac{\sqrt{11}}{6}$ gives vertical component
- **Method 3**: Verify with one specific segment using Pythagorean theorem

D. Answer Choice Validation

- Our answer: $\arccos \frac{5}{6}$
- Matches choice: **(A)**
- Why not the others?
  - (B) $\arccos \frac{4}{5}$: Would give $100/120 = 4/5$, but that's $0.8 \neq 5/6 \approx 0.833$
  - (C) $\arccos \frac{3}{10}$: Would need $\cos\theta = 0.3$, too small (very large angle)
  - (D) $\arcsin \frac{4}{5}$: Uses arcsin instead of arccos
  - (E) $\arcsin \frac{5}{6}$: Uses arcsin instead of arccos

Solution Roadmap

6. Solution Roadmap

A. High-Level Solution Path

1. **Recognize pattern**: Horizontal distance 100 m, path length 120 m
2. **Identify key ratio**: $\frac{100}{120} = \frac{5}{6}$
3. **Connect to trigonometry**: This ratio represents $\cos\theta$
4. **Solve for angle**: $\theta = \arccos \frac{5}{6}$
5. **Select answer**: Choice (A)

B. Detailed Solution Steps

Step 1: Understand the geometry.

- Rectangle is 100 m $\times$ 30 m

- Zigzag path starts at A, ends somewhere on $\overline{BC}$
- Total path length is 120 m
- All angles θ are equal

Step 2: Use the unfolding technique.

- Imagine "unfolding" the zigzag path by reflecting segments
- When unfolded, the path becomes a straight line
- This straight line has length 120 m
- The horizontal distance covered is 100 m (rectangle length)

Step 3: Apply trigonometry.

- The unfolded path forms the hypotenuse of a right triangle
- The horizontal distance is the adjacent side
- Therefore: $\cos\theta = \frac{\text{horizontal}}{\text{path}} = \frac{100}{120} = \frac{5}{6}$

Step 4: Solve for the angle.

- $\theta = \arccos\frac{5}{6}$

Step 5: Select the answer.

- This matches choice **(A)**

C. Key Decision Points

- **Decision 1**: Recognize that horizontal distance / path length gives $\cos\theta$
- **Decision 2**: Use unfolding technique (or similar approach)
- **Decision 3**: Identify answer as arccos, not arcsin

D. Time Management

- Understanding problem: 30 seconds
- Computing ratio $100/120 = 5/6$: 20 seconds
- Connecting to $\cos\theta$: 45 seconds
- Solving for θ: 30 seconds
- Verification: 45 seconds
- **Total**: 3 minutes 30 seconds

Competition Strategy

7. Competition Strategy Integration

A. Problem Triage

- **Difficulty Assessment**: Medium (Problem 15/25)
- **Confidence Level**: High if you recognize the unfolding technique
- **Decision**: Attempt if comfortable with geometry/trig; otherwise return later

B. Time Allocation

- **Target Time**: 3-4 minutes
- **Maximum Time**: 5 minutes
- **Quick Approach**: If you see $100 : 120 = 5 : 6$ immediately, guess **(A)** $\arccos \frac{5}{6}$

C. Solution Format

- Multiple choice: Select **(A)** $\arccos \frac{5}{6}$
- Verification: Check ratio and answer format
- Bubble answer sheet carefully

D. Strategic Considerations

- **Pattern Recognition**: If you've seen unfolding problems before, this is quick
- **Elimination**: Can eliminate (D) and (E) if you determine answer uses arccos
- **Ratio Clue**: $5 : 6$ ratio strongly suggests answer (A)
- **Time Saver**: Don't overthink - the elegant solution is usually correct

Common Pitfalls and Warnings

Common Pitfalls and Warnings

A. Conceptual Errors

- **Confusing** arcsin **with** arccos: The horizontal component uses cosine, not sine
- **Using wrong ratio**: Don't use $\frac{30}{120}$ or $\frac{30}{100}$ - these don't directly give the angle
- **Forgetting unfolding**: Trying to set up complex equations instead of using elegant unfolding
- **Assuming integer segments**: The number of zigzag segments doesn't need to be an integer

B. Computational Errors

- **Ratio simplification**: $\frac{100}{120} = \frac{5}{6}$, not $\frac{4}{5}$ or other values
- **Angle type**: Answer is arccos, not arcsin (check which trig function you're inverting)
- **Arithmetic**: $100 \div 120 = 5 \div 6$, verify this carefully

C. Strategic Errors

- **Overcomplicating**: Don't set up unnecessary variables or equations
- **Ignoring answer choices**: The choices tell you the answer involves $\frac{5}{6}$, $\frac{4}{5}$, or $\frac{3}{10}$
- **Time management**: If stuck after 3 minutes, guess (A) and move on

D. Warning Signs

- If you get $\sin\theta = \frac{5}{6}$, reconsider - horizontal component is cosine
- If you get $\cos\theta = \frac{4}{5}$, recheck your ratio calculation
- If your answer isn't among the choices, revisit the unfolding concept

Detailed Solution

A. Main Solution (Unfolding Method)

1. **Visualize the unfolding**:
   - The zigzag path reflects between top and bottom edges
   - Imagine "unfolding" by reflecting alternating segments
   - The unfolded path becomes a straight line from A to endpoint
2. **Identify distances**:
   - Horizontal distance (rectangle length): 100 m
   - Total path length (unfolded hypotenuse): 120 m
3. **Apply trigonometry**:

$$\begin{aligned}\cos\theta &= \frac{\text{adjacent}}{\text{hypotenuse}} \\ &= \frac{\text{horizontal distance}}{\text{path length}} \\ &= \frac{100}{120} \\ &= \frac{5}{6}\end{aligned}$$

4. **Solve for angle**:

$$\theta = \arccos\frac{5}{6}$$

B. Alternative Solution (Similar Triangles)

1. Let x be the horizontal projection of one zigzag segment
2. The vertical component is 30 m (width of rectangle)
3. By Pythagorean theorem, segment length $= \sqrt{x^2 + 900}$
4. All segments are similar (same angle θ)
5. Total horizontal distance: 100 m
6. Total path length: 120 m
7. Since all segments are similar with the same ratio:

$$\frac{\text{total horizontal}}{\text{total path}} = \frac{x}{\sqrt{x^2 + 900}}$$

8. This gives us:

$$\frac{100}{120} = \cos\theta = \frac{5}{6}$$

9. Therefore: $\theta = \arccos\frac{5}{6}$

C. Physics Interpretation

- Imagine Usain walks at constant speed
- The horizontal velocity component is constant (like light reflecting)
- Horizontal velocity = (total velocity) $\times \cos\theta$
- Since time is the same, distance ratio equals velocity ratio:

$$\frac{\text{horizontal distance}}{\text{total distance}} = \cos\theta = \frac{100}{120} = \frac{5}{6}$$

- Therefore: $\theta = \arccos\frac{5}{6}$

D. Verification

- Check ratio: $\frac{100}{120} = \frac{5}{6}$ ✓
- Verify answer type: arccos (not arcsin) ✓
- Matches choice (A): $\arccos\frac{5}{6}$ ✓
- Reasonableness: $\theta \approx 33.6°$, appropriate for zigzag ✓

Key Insights

Key Insight: The most elegant approach is the **unfolding technique**. When you "unfold" a zigzag path by reflecting alternating segments, the path becomes a straight line. The horizontal distance remains 100 m, and the total path length is 120 m. This creates a right triangle where:

$$\cos\theta = \frac{\text{horizontal distance}}{\text{path length}} = \frac{100}{120} = \frac{5}{6}$$

Therefore: $\theta = \arccos\frac{5}{6}$

General Pattern: For zigzag problems:

- Horizontal progress = (path length) $\times \cos\theta$
- Vertical progress = (path length) $\times \sin\theta$
- Unfolding preserves lengths and reveals the geometry

Physics Analogy: Like light reflecting between parallel mirrors, the horizontal velocity component remains constant. This is why the ratio of horizontal distance to total path length equals $\cos\theta$.

Why this works: The unfolding technique works because:

- Each reflection preserves the angle
- The sum of horizontal projections equals the rectangle length
- The sum of segment lengths equals the total path length
- These ratios are preserved in the unfolded configuration

Final Answer

The angle θ that produces a path length of 120 meters is:

$$\boxed{\textbf{(A)}\ \arccos\frac{5}{6}}$$

Solution Summary:

- Horizontal distance: 100 m
- Path length: 120 m
- Using unfolding: $\cos\theta = \frac{100}{120} = \frac{5}{6}$
- Therefore: $\theta = \arccos\frac{5}{6}$

Confidence Assessment

- **Confidence Level**: 99%
  - Elegant unfolding technique gives clear answer
  - Simple ratio calculation: $100/120 = 5/6$
  - Multiple verification methods confirm result
  - Answer format matches problem type
- **Verification Applied**: Level 3 - Reasonableness + Alternative Method
  - Checked ratio calculation
  - Verified answer type (arccos vs arcsin)
  - Confirmed with physics interpretation
  - All methods agree
- **Time-Confidence Trade-off**: High confidence achieved in 3-4 minutes